# Chapter 4 Notes

## Section 1

### Notes

Note that when Smullyan says: "Thus, for example, the class $V$ is partially ordered under inclusion," by inclusion he means *set inclusion* (that is, being a subset), not *set membership*.

Here is the proof of part (2) of Proposition 1.3. For suppose that some element $x$ had two predecessors $y$ and $z$. Then $y < x$ and $z < x$. If $y < z$, then $y < z < x$, contrary to the fact that $y$ is the predecessor of $x$. Similarly it cannot be that $z < y$. But, because $\leq$ is a linear ordering, either $y \leq z$ or $z \leq y$. In either case, because neither $y < z$ or $z < y$ hold, $y = z$, thus completing the proof.

Note that, for a linear ordering $\leq$ of $A$, all finite non-empty subclasses of $A$ must have a least element. We prove this by induction on the size of the subclass. The base case is obvious (because, by $O_1$, $x \leq x$ for all $x$ in the field of $\leq$). Now suppose the result holds for some $n$. Consider some subclass $S$ of $A$ of size $n+1$. Then for any element $x \in S$, suppose $x$ is not the least element. Then $S - \{x\}$, by the induction assumption, has a least element $a$. Because $x$ is not the least element, $a \leq x$ (by $O_4$), meaning $a$ is the least element of $S$. Thus, for every element $x \in S$, either $x$ is the least element of $S$, or there is some other element least element $a$ of $S$. In any case, $S$ has a least element.

Here is an explicit proof of Proposition 1.7: We can prove by induction that, for a well-ordering $\leq$, every non-empty finite subclass of the field of $\leq$ has a greatest element. The base case, when the subclass has one element, is obvious (because, by $O_1$, $x \leq x$ for all $x$ in the field of $\leq$). Now suppose that, for some non-empty subclass of size $n$, it has a greatest element. Now consider some non-empty subclass $C$ of size $n+1$. By $O_5$, it has a least element $e$. Then $C - \{e\}$ is of size $n$, and thus by the induction assumption, has a greatest element $a$. But because $e$ is the least element of $C$, $e \leq a$. Hence, for all elements $x$ of $C$, $x \leq a$, meaning $a$ is the greatest element of $C$. This completes the induction.

Now having proven this result, suppose that $x$ was a limit element of a well ordering $\leq$ and $L_<(x)$ had finitely many elements. Then, by the above result, $L_<(x)$ must have a greatest element, and thus, by Proposition 1.6, $x$ would not be a limit element, a contradiction.

### Exercise 1.1

Suppose that $\leq$ is a partial ordering such that every non-empty subclass of the field of $\leq$ has a least element. We need to show that, for every $x$ and $y$ in the field of $\leq$, $x \leq y \vee y \leq x$. If $x$ and $y$ are both in the field of $\leq$, then we know that the class $\{x, y\}$ has a least element. If $x$ is the least element, then $x \leq y$, and if $y$ is the least element, then $y \leq x$. Thus either $x \leq y$ or $y \leq x$, therefore completing the proof.

### Exercise 1.2

I do not think the premise of this problem makes sense. For suppose that a linear ordering $\leq$ of $A$ was such that, for every proper lower section $L$ of $A$, there is a least element $x$ of $A$ not in $L$. In fact, we can then prove that $A$ has no proper lower sections. For suppose it has some proper lower section $L$. Then there would be a least element $x$ of $A$ not in $L$. Thus, for every element $y \in L$, $y \leq x$. But, by the definition of a proper lower section, also $x < y$, which contradicts $O_2$. Hence

$A$ can have no proper lower sections. Then obviously $\leq$ cannot be a well-ordering of $A$. For if it was, then $A$ would have a least element $e$. But then $\{e\}$ would be a proper lower section of $A$, contrary to the fact that $A$ has no proper lower sections. Thus $\leq$ cannot be a well-ordering of $A$.

## Exercise 1.3

This problem had similar issues as in Exercise 1.2: assumption (c) of the problem implies that there can be no proper lower sections $L$ of $A$ such that $L$ has no greatest element.

## Exercise 1.4

We prove that in a well ordering, all backward paths must be finite. What this amounts to saying is that, if $\leq$ is a well-ordering, then there can be no denumerable backward paths. So suppose there was some denumerable backwards path $a_0, a_1, ..., a_n, ....$ Because this is a well-ordering, the class $\{a_0, a_1, ..., a_n, ...\}$ has a least element. Thus for some $m$, it must be the case that $a_m$ is the least element of $\{a_0, a_1, ..., a_n, ...\}$. Because the sequence is denumerable, there must be a term $a_{m+1}$ of the sequence. By the definition of a backward path, $a_{m+1} < a_m$. But also $a_m \leq a_{m+1}$, which is a contradiction (by $O_2$, as this would imply $a_m = a_{m+1}$, but because $a_{m+1} < a_m$, $a_m \neq a_{m+1}$).

## Exercise 1.5

(a): On the one hand, we assume the first form of the AC. Now consider any non-empty set $A$ of pairwise disjoint non-empty sets. By the AC, there is some function $f$ such that $f(x) \in x$ for all elements $x \in A$. Then take $C$ to be the class of all $x$ such that $x = f(y)$ for some $y \in A$. Because $A$ is a set of pairwise disjoint sets, and $f(y) \in y$ for every $y \in A$, it follows that $C$ contains exactly one element of each element of $A$.

(b): On the other hand, suppose that the second form of the AC holds. Now consider any non-empty set $A$ of non-empty sets. Note that for any set $x$, $\{x\}$ is a non-empty set of pairwise disjoint non-empty sets, and hence, there exists some set $C$ that consists of exactly one element of $x$. Denote this set $C$ for this set $x$ as $C_x$, and let $a_x$ be the one element of $C_x$. Then take the function $f$ to be the set of all ordered pairs $\langle x, a_x \rangle$, which is clearly a function such that for each $x$ in $A$, $f(x) \in x$.

## Exercise 1.6

It suffices to show that this form is equivalent to the first form, and then, by Exercise 1.5, it follows that it is also equivalent to the form in that problem.

(a): On the one hand, suppose that the first form of the AC holds, and consider some non-empty set $A$ of non-empty sets. Then, by the AC, there is some function $f$ such that $f(x) \in x$ for all elements $x$ of $\mathcal{P}(A) - \{A\}$ (that is, the set of all proper subsets of $A$), that is, for all proper subsets $x$ of $A$, $f(x) \in x$.

(b): Now suppose that the second form (the one presented in this problem) of the AC holds, and consider some non-empty set $A$ of non-empty sets. Then $\cup A$ is a set, and hence, by the AC, there is some function $f$ such that $f(x) \in x$ for all proper subsets of $\cup A$. Then we take the function $g$ to be the set of all ordered pairs $\langle x, f(x) \rangle$ for all elements $x$ of $A$, which works, as $x$ is a subset of $\cup A$.

## Exercise 1.7

We wish to prove that every infinite set can be put into a 1-1 correspondence with itself (assuming the AC). So, consider any infinite set $A$. Then, by exercise 1.8, it has a denumerable subset $D$, and there is a 1-1 function $f$ from the natural numbers to elements of $D$. Pick some element of $D$ ($f(0)$, say). Then $D - f(0)$ is still a denumerable subset. Now we can define a 1-1 function $g$ from $A$ to $A - \{f(0)\}$ (a proper subset of $A$) as follows: Assume $x \in A$. If $x \in D$, then $x = f(n)$

for some $n$, and we add the ordered pair $\langle f(n), f(n^+)\rangle$ to the function $g$. If $x \notin D$, then we add the ordered pair $\langle x, x\rangle$ to $g$. Thus this a 1-1 function $g$ from $A$ onto $A - \{f(0)\}$, therefore completing the proof.

### Exercise 1.8

It can easily be shown by induction that every infinite set $A$ contains an infinite amount of subsets of size $n$ for every $n > 0$. The base case, when $n = 1$, holds, as, because $A$ is infinite, there is, for every element $x$ of $A$, the subset $\{x\}$ of $A$, and there are infinitely many of these. Now suppose the result holds for some $n$, that is, $A$ has an infinite amount of subsets of size $n$. For each of these subsets $S$ of size $n$, $A - S$ (which must also be infinite) has an infinite number of subsets of size 1 (by having proven the base case), and then, for each element $x \in A - S$, the subset $S \cup \{x\}$ is of size $n + 1$, and because there are infinitely many of these (because $A - S$ contains infinitely many elements), the induction is complete.

Now consider any infinite set $A$. Then the class $C$ of all subsets of $A$ of size one is a set (because it is a subset of $\mathcal{P}(A)$). Then $\{C\}$ is a set, meaning, by the AC, there is some function $f_0^0$ such that $f_0^0(C) \in C$, that is, $f_0^0(C)$ is a subset of $A$ (that does not include the empty set) of size one. Because $f_0^0(C)$ is a set, so is $\{f_0^0(C)\}$ (which is a non-empty set of non-empty sets), meaning, by the AC, there is some function $f_1^0$ such that $f_1^0(f_0^0(C)) \in f_0^0(C)$, that is, $f_1^0(f_0^0(C))$ is an element of $A$ ($f_1^0(f_0^0(C))$ is an element of a subset of $A$ of size one). Then we can establish a denumerable sequence $a_0, a_1, ..., a_n, ...$ of elements of $A$ as follows: Take $a_0$ to be $f_1^0(f_0^0(C))$, and let $S_0 = A$ and $C_0 = C$. Then let $S_1 = S_0 - \{a_0\}$ and $C_1$ be the set of all subsets of $S_1$ of size one, and, in a similar manner as in how we obtained $f_1^0(f_0^0(C))$, we can define functions $f_1^1$ and $f_0^1$ using the AC such that $f_1^1(f_0^1(C_1))$ is an element of $S_1$ not equal to $a_0$ (because $S_1$ does not contain $a_1$). Then take $a_1$ to be $f_1^1(f_0^1(C_1))$. In general, if we have $a_i$ and $S_i$ for all numbers $i$ less than $n$, we take $S_{n+1}$ to be $S_n - \{a_n\}$, $C_{n+1}$ to be the set of all subsets of $S_{n+1}$ of size one, and we can define functions $f_1^{n+1}$ and $f_0^{n+1}$ using the AC such that $f_1^{n+1}(f_0^{n+1}(C_{n+1}))$ is an element of $S_{n+1}$ not equal to $a_i$ for all $i \leq n$. Obviously, each $a_i$ is distinct from all others in the sequence. Thus the set of all $a_i$ is a denumerable subset of $A$.

## Section 2

### Notes

Note that $\omega$ is not superinductive (under the successor function). This is because it is not the case that $\omega$ is closed under chain unions. For instance, consider the subset $C : \{1, 2, ..., n, ...\}$ of $\omega$, which is obviously a chain. We show that it is not the case that $\cup C \in \omega$. For suppose $\cup C \in \omega$. Then, by definition of $\omega$, $\cup C$ would be some natural number $n$. Obviously $n \neq 0$. Now, $n^+ \notin n$ (by Theorem 5.6 of Chapter 3). But $n^{++}$, which includes $n^+$, is an element of $C$; thus $n^+$ is an element of $\cup C$, a contradiction. Hence $\cup C \notin \omega$.

Here is an explanation of why $M$ is minimally superinductive under $g$ in Theorem 2.8. We know that $M$ is the intersection of all subsets of $S$ that are superinductive under $g$. Now, let $C$ be the set of all subsets of $S$ that are superinductive under $g$. Then $M = \cap C$. Because every element of $C$ is superinductive under $g$, every element of $C$ is closed under $g$, and every element of $g$ is closed under chain unions. Thus, by Lemma 2.7, $\cap C$, that is, $M$, is closed under $g$ and is closed under chain unions. Obviously $\emptyset \in M$ (because $\emptyset$ is an element of all superinductive sets, and hence of all elements of $C$). Lastly, we must show that $M$ is minimally superinductive under $g$. Suppose that some proper subset $A$ of $M$ was superinductive under $g$. Thus there must be some element $a \in M$ such that $a \notin A$. Now, clearly $A$ must be a subset of $S$. Thus $A$ is a subset of $S$ which is superinductive under $g$. But then $A \in C$, and because $a \in M$ ($a \in \cap C$), $a \in A$, a contradiction. Thus $M$ must be minimally superinductive under $g$ (and $M$ is a set because $M \subseteq S$).

Here is why $\cup M \in S$ in the proof of Theorem 2.8. As remarked in the proof, we know that $\cup M \in M$. But $M$ is the intersection of all subsets of $S$ that are superinductive under $g$. But because $S$ itself is superinductive under $g$ (and is a subset of itself), $\cup M \in M$ implies that $\cup M \in S$.

There are multiple things to be noted for the proof of Theorem 2.10. First, the definition of $S_b$ is not correct. For, given the current definition, we cannot definitively know that $S_b$ is closed under $g'$. For suppose that $x \in S_b$. Then either $x \neq \emptyset$ or $x = \emptyset$. In the first case, we know that $g'(x) = g(x)$, and because $x \subseteq g(x)$ (because $g$ is progressing), $x \subseteq g'(x)$. Thus, because $x$ is a set which includes $b$ (by the definition of $S_b$), $g'(x)$ also includes $b$, meaning $g'(x) \in S_b$. However, if $x = \emptyset$, then $g'(x) = b$. But there is no guarantee that $b \in S_b$: it very well could be the case that $b \notin b$. In fact, it very well could be that $S_b$ (defined in this way) only contains the empty set!

I will remark that this (incorrect) definition of $S_b$ does make $S_b$ closed under chain unions. For consider any chain $C$ of elements of $S_b$. Then consider $\cup C$. Each element of $C$ is either $\emptyset$ or a set $x$ such that $b \in x$. If $C$ only contains $\emptyset$, then $\cup C = \emptyset$, and thus $\cup C \in S_b$. Otherwise, $C$ contains some element $x$ such that $b \in x$. But then $\cup C$ contains all elements of $x$, and hence contains $b$, meaning $b \in \cup C$, and thus $\cup C \in S_b$.

The proof of Theorem 2.10 which I found is very similar to the one given in the book, just with a slightly different definition of $S_b$:

**Corrected Proof of Theorem 2.10**:
Assume the hypothesis. As in the original proof, the case when $b = \emptyset$ is just Thoerem 2.8; so assume $b \neq \emptyset$. We let $S_b$ be the set of all elements $x$ of $S$ for which $b \subseteq x$ (and this *does* include $b$), together with $\emptyset$. This is closed under chain unions for the following reason: Consider some chain $C$ of elements of $S_b$. Then consider $\cup C$. Each element of $C$ is either $\emptyset$ or a set that $b$ is a subset of. If $C$ only contains $\emptyset$, then $\cup C = \emptyset$, and thus $\cup C \in S_b$. Otherwise, $C$ contains some element $x$ such that $b \subseteq x$. But then $\cup C$ contains all of the elements of $x$, and thus also all of the elements of $b$, meaning $b \subseteq \cup C$, and thus $\cup C \in S_b$. Thus $S_b$ is closed under chain unions. Then we define $g'$ in the same way as in the book: $g'(\emptyset) = b$, and for any $x \in S_b$ other than $\emptyset$, take $g'(x) = g(x)$. Then $g'$ is progressing for the following reason: For consider any $x \in S_b$. If $x = \emptyset$, then obviously $x \subseteq g'(x)$. Otherwise, if $x \neq \emptyset$, then $g'(x) = g(x)$, and because $x \subseteq g(x)$ (because we assumed that $g$ was progressing), $x \subseteq g'(x)$. Thus $g'$ is progressing. Next, we need to show that $S_b$ is closed under $g'$. For suppose that $x \in S_b$. If $x = \emptyset$, then $g'(x) = b$, and because $b \in S_b$, $g'(x) \in b$. Now suppose $x \neq \emptyset$. Because $x \in S_b$ (and $x \neq \emptyset$), $b \subseteq x$. Thus, because $x \subseteq g'(x)$, $b \subseteq g'(x)$, meaning (by the definition of $S_b$), $g'(x) \in S_b$. Thus $S_b$ is closed under $g'$. Lastly, here is why $S_b$ is a set: $S_b - \{\emptyset\}$ is a set because it is a subset of $S$ (which is assumed to be a set), and thus $\cup\{(S_b - \{\emptyset\}), \{\emptyset\}\}$, that is, $S_b$, is also a set. Thus $S_b$ satisfies the criteria of Theorem 2.8, meaning there is some element $x$ of $S_b$ such that $g'(x) = x$. Now, it cannot be that $x = \emptyset$, or else $b = g'(x) = x = \emptyset$. Thus $x \neq \emptyset$, meaning (by the definition of $S_b$) $b \subseteq x$. Also, because $x \neq \emptyset$, $g'(x) = g(x)$, so that $g(x) = x$. Thus $b \subseteq x$ and $g(x) = x$, therefore completing the proof (and obviously $x \in S$).

It very well could be that the book simply contained a typo, and meant to define $S_b$ in the way it was defined in the correct proof.

## Exercise 2.1

Consider any set $S$:

(1) Suppose that every element of $S$ is closed under a function $g$. Now consider $\cap S$. To prove that $\cap S$ is also closed under $g$, we need to show that if $x \in \cap S$, then $g(x) \in \cap S$. So suppose $x \in \cap S$. Then $x$ is an element of every element of $S$. But every element of $S$ is closed under $g$. Thus $g(x)$ is an element of every element of $S$, meaning $g(x) \in \cap S$.

(2) Suppose that every element of $S$ is closed under chain unions. Now consider $\cap S$. To prove that $\cap S$ is also closed under chain unions, we need to show that for any chain $C$ of elements of $\cap S$, $\cup C \in A$. So consider some chain $C$ of elements of $\cap S$. Thus $C$ is also a chain of elements of every element of $S$. But, because every element of $S$ is closed under chain unions, $\cup C$ is an element of every element of $S$. Thus $\cup C \in \cap S$.